OpenShift 4.20 IPI Install on vSphere — Airgapped, No External LB

Runbook — 3 control plane + 2 compute, DHCP + Mirror Registry already in place

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0. Environment assumptions (edit before use)

Item	Value (example — replace)
Cluster name	`ocp1`
Base domain	`example.local`
FQDN of cluster	`ocp1.example.local`
vCenter	`vcenter.example.local`
Datacenter / Cluster / Datastore	`DC1` / `Cluster1` / `ds-ocp01`
VM Network (port group)	`PG-OCP-NODES`
Machine network CIDR	`10.10.10.0/24`
API VIP	`10.10.10.5`
Ingress VIP	`10.10.10.6`
Mirror registry	`mirror.example.local:8443`
DNS servers	`10.10.10.2`
NTP servers	`10.10.10.2`
DHCP range for nodes	already configured (per you)

Why no external LB works here: vSphere IPI does **not** require an external LB. When you set `apiVIPs` / `ingressVIPs` in `install-config.yaml`, the installer deploys **keepalived** as static pods on the control-plane nodes (managed by the Machine Config Operator). Keepalived owns the API VIP (always pinned to a healthy control-plane node) and the Ingress VIP (pinned to a node running a router pod), using VRRP. `api-int` does not need a VIP or LB at all — it resolves to the same API VIP.

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1. DNS records required

Create these **before** starting the install. All must resolve from the bastion/mirror host, the vCenter network, and (after install) from wherever `oc` /browsers will be used.

Record	Type	Value	Notes
`api.ocp1.example.local`	A	`10.10.10.5` (API VIP)	External + internal API access
`api-int.ocp1.example.local`	A	`10.10.10.5` (API VIP)	Internal API, same VIP — **no separate LB needed**
`*.apps.ocp1.example.local`	A (wildcard)	`10.10.10.6` (Ingress VIP)	Routes/Ingress for all apps
`mirror.example.local`	A	mirror registry VM IP	Must already resolve — you said mirror registry is in place
`vcenter.example.local`	A	vCenter IP	Installer talks to vCenter API
Reverse (PTR) for VIPs	PTR	optional but recommended	Some checks are happier with clean reverse lookups

> Bootstrap/master/worker node hostnames do **not** need pre-created DNS records — DHCP + the cluster's internal CoreDNS + `/etc/hosts` handle node-to-node resolution automatically in IPI. Only `api`, `api-int`, and `*.apps` are mandatory external records.

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2. vSphere networking prerequisite (commonly missed — causes keepalived/VIP failures)

Because there's no external LB, the VIPs are brought up as **secondary IPs / MAC address failover** on the control-plane vNICs via keepalived (VRRP). ESXi blocks this by default. On the **port group** (`PG-OCP-NODES`) used by the cluster VMs, set:

vSphere Networking Security Setting	Required Value
Promiscuous mode	**Accept**
MAC Address Changes	**Accept**
Forged Transmits	**Accept**

Without this, the API/Ingress VIP will never come up or will flap, and `bootstrap-complete` will hang.

Also confirm:

- The port group allows the full node + VIP subnet (no private VLAN restrictions).
- vCenter account used by the installer has the standard IPI required privileges (Datastore, Resource, Network, VM provisioning, Folder, etc.) — required regardless of LB choice.

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3. NSX-T DFW / firewall rules

Apply these on the NSX-T Distributed Firewall (or edge FW) between the relevant zones. Source/Dest groups are illustrative — map to your NSX groups.

#	Source	Destination	Port/Proto	Purpose
1	Bastion/mirror host	vCenter	tcp/443	Installer → vCenter API
2	All OCP nodes (bootstrap+master+worker)	vCenter	tcp/443	vSphere cloud provider / CSI
3	All OCP nodes	Mirror registry (`mirror.example.local`)	tcp/8443 (or your mirror port)	Pull release + operator images
4	All OCP nodes	DNS servers	tcp+udp/53	Name resolution
5	All OCP nodes	NTP servers	udp/123	Time sync (etcd is time-sensitive)
6	Bastion	All OCP nodes	tcp/22	SSH troubleshooting (core user)
7	Client/Admin subnets	API VIP	tcp/6443	`oc` /console access to API
8	Client/Admin subnets	Ingress VIP	tcp/443, tcp/80	Routes/console/apps
9	Node ↔ Node (control-plane + worker, all)	Node ↔ Node	tcp/6443	Kubernetes API (internal)
10	Node ↔ Node	Node ↔ Node	tcp/22623	Machine Config Server
11	Node ↔ Node	Node ↔ Node	tcp/2379-2380	etcd (control-plane only, but allow cluster-wide subnet)
12	Node ↔ Node	Node ↔ Node	tcp/10250-10259	Kubelet, scheduler, controller-manager
				OVN-Kubernetes Geneve overlay

13	Node ↔ Node	Node ↔ Node	udp/6081	(default network type)
14	Node ↔ Node	Node ↔ Node	tcp+udp/9000-9999	Host-level cluster services (node exporter etc.)
15	Node ↔ Node	Node ↔ Node	tcp+udp/30000-32767	NodePort services range
16	Control-plane ↔ Control-plane	Control-plane ↔ Control-plane	**IP protocol 112 (VRRP)**	Keepalived VIP failover **— critical, easy to miss on NSX since it's not TCP/UDP**
17	Node ↔ Node	Node ↔ Node	icmp	VRRP/keepalived + general health checks recommend allowing ICMP echo

> Rule 16 is the one most people forget on NSX because the DFW UI defaults to TCP/UDP service objects — you need to explicitly create an IP-protocol service object for **protocol 112 (VRRP)** and allow it between control-plane node IPs (and ideally the VIP itself).

Also confirm **East-West is not blocked between node subnet and VIP subnet** if they differ, and that the mirror registry's self-signed/internal CA is trusted (see `additionalTrustBundle` below) rather than solved by opening extra ports.

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4. Pre-flight checklist

- DNS records from §1 created and resolving from bastion
- Port group security settings from §2 applied
- NSX DFW rules from §3 applied (incl. protocol 112)
- Mirror registry reachable on 443/8443 from bastion and (post-boot) from nodes' future subnet
- Mirror registry CA certificate exported (for `additionalTrustBundle`)
- `oc-mirror` / mirrored release payload + operator catalog already pushed to mirror registry, and `imageContentSourceContexts` /results manifests from the mirror run saved
- Pull secret (merged with mirror registry auth) available
- SSH key pair generated for `core` user access

- vCenter service account with IPI-required privileges created
- `openshift-install` , `oc` , `kubectl` binaries extracted from the **mirrored** release image (not from the internet) onto the bastion

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5. Extract installer & oc from the mirror (airgapped-correct way)

```
# On the bastion (which has access to the mirror registry only)

# Pull secret must include the mirror registry auth
export PULL_SECRET_FILE=~/pull-secret-mirror.json

# Get the exact release image digest you mirrored
export LOCAL_REGISTRY='mirror.example.local:8443'
export LOCAL_REPO='ocp4/openshift4'
export OCP_RELEASE=4.20.0   # match what you mirrored

# Extract the CLI (oc, kubectl) from the mirrored release
oc adm release extract \
  -a ${PULL_SECRET_FILE} \
  --command=oc \
  --to=/usr/local/bin \
  ${LOCAL_REGISTRY}/${LOCAL_REPO}:${OCP_RELEASE}-x86_64

# Extract the installer binary from the mirrored release
oc adm release extract \
  -a ${PULL_SECRET_FILE} \
  --command=openshift-install \
  --to=/usr/local/bin \
  ${LOCAL_REGISTRY}/${LOCAL_REPO}:${OCP_RELEASE}-x86_64

openshift-install version
```

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6. Gold-standard `install-config.yaml` (fully commented)

Save as `~/ocp-install/install-config.yaml` . **Back it up** — the installer consumes/deletes it during `create manifests` .

```
apiVersion: v1
baseDomain: example.local          # your base domain (NOT including cluster name)
metadata:
  name: ocp1                       # cluster name -> becomes ocp1.example.local

# ---- Compute (worker) pool ----
compute:
```

```
- name: worker
  replicas: 2                      # matches your 2-compute topology
  platform:
    vsphere: {}                    # per-pool overrides not needed for a single-DC instal

# ---- Control plane pool ----
controlPlane:
  name: master
  replicas: 3                      # 3 control-plane nodes

# ---- Networking ----
networking:
  networkType: OVNKubernetes       # default/required overlay for 4.20 (needs udp/6081 op
  clusterNetwork:
  - cidr: 10.128.0.0/14
    hostPrefix: 23
  serviceNetwork:
  - 172.30.0.0/16
  machineNetwork:
  - cidr: 10.10.10.0/24            # the subnet your DHCP hands out node IPs on; VIPs mus

# ---- vSphere platform ----
platform:
  vsphere:
    apiVIPs:
      - 10.10.10.5                 # <-- No external LB: keepalived owns this VIP
    ingressVIPs:
      - 10.10.10.6                 # <-- No external LB: keepalived owns this VIP
    vcenters:
    - server: vcenter.example.local
      user: svc-ocp-installer@vsphere.local
      password: '<VCENTER_PASSWORD>'
      datacenters:
      - DC1
    failureDomains:
    - name: dc1-cluster1
      region: region1              # arbitrary label, must match a tag/zone scheme if you
      zone: zone1
      server: vcenter.example.local
      topology:
        datacenter: DC1
        computeCluster: /DC1/host/Cluster1
        networks:
        - PG-OCP-NODES              # the port group with promiscuous/forged-transmit/mac
        datastore: /DC1/datastore/ds-ocp01
        resourcePool: /DC1/host/Cluster1/Resources   # optional, omit to use root RP
        folder: /DC1/vm/ocp1                          # optional VM folder

# ---- Airgapped / mirror registry settings ----
imageDigestSources:                 # 4.13+ name for what used to be imageContentSources
- source: quay.io/openshift-release-dev/ocp-release
```

```
  mirrors:
  - mirror.example.local:8443/ocp4/openshift4
- source: quay.io/openshift-release-dev/ocp-v4.0-art-dev
  mirrors:
  - mirror.example.local:8443/ocp4/openshift4
# ^ Use the EXACT mirror list produced by your oc-mirror / oc adm release mirror run —
#   copy-paste from ImageContentSourcePolicy.yaml / mirror-results, do not hand-type.

additionalTrustBundlePolicy: Always  # trust the bundle below even without an explicit pro
additionalTrustBundle: |
  -----BEGIN CERTIFICATE-----
  <PASTE MIRROR REGISTRY CA CERT HERE>
  -----END CERTIFICATE-----

# ---- Auth ----
pullSecret: '{"auths":{"mirror.example.local:8443":{"auth":"<BASE64_USER_PASS>","email":"
sshKey: 'ssh-ed25519 AAAA... core@bastion'

fips: false
publish: Internal        # airgapped best practice: do not attempt to publish external DNS
```

Notes

- `publish: Internal` prevents the installer from trying to reach any external DNS provider (irrelevant on-prem, but is the documented safe default for disconnected on-prem installs).
- Do **not** set `platform.vsphere.loadBalancer.type: UserManaged` — that field is only for when you *do* have an external LB. Leaving it unset (or `OpenShiftManagedDefault`, the default) is what makes the installer deploy keepalived.
- Keep a copy of this file — `create manifests` / `create cluster` consumes it.

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7. Generate manifests, ignition, and install

```
mkdir ~/ocp-install-run && cp ~/ocp-install/install-config.yaml ~/ocp-install-run/

# Generate manifests (review only if you need custom MachineConfigs, e.g. NTP)
openshift-install --dir=~/ocp-install-run create manifests

# --- OPTIONAL: drop in extra manifests here, e.g. chrony NTP MachineConfig (see §8) ---

# Kick off the install (creates ignition configs + provisions VMs via vCenter API)
openshift-install --dir=~/ocp-install-run create cluster --log-level=info
```

Watch for:

- `Bootstrap-complete` — bootstrap node has handed off to control plane; safe to delete bootstrap VM after this (installer does it automatically unless `--log-level=debug` interrupted).

- If it hangs at bootstrap, the two most common airgapped/no-LB causes are: (a) nodes can't reach the mirror registry (check NSX rule #3 + CA trust), or (b) VIP never comes up (check §2 port group settings + NSX protocol 112).

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8. Optional gold-standard extra manifest: NTP via chrony (recommended, etcd needs accurate time)

Drop this into `~/ocp-install-run/openshift/99_masters-chrony-configuration.yaml` **before** `create cluster` (or apply as a MachineConfig post-install). Duplicate with `role: worker` for a worker version if needed.

```
apiVersion: machineconfiguration.openshift.io/v1
kind: MachineConfig
metadata:
  labels:
    machineconfiguration.openshift.io/role: master
  name: 99-masters-chrony-configuration
spec:
  config:
    ignition:
      version: 3.2.0
    storage:
      files:
        - contents:
            source: data:text/plain;charset=utf-8;base64,c2VydmVyIDEwLjEwLjEwLjIgaWJ1cnN0(
            # decoded: server 10.10.10.2 iburst / drift/rtcsync/logdir defaults
          mode: 420
          overwrite: true
          path: /etc/chrony.conf
```

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9. Post-install: mirrored OperatorHub CatalogSource

Since the whole `redhat-operator-index` was mirrored (per your earlier plan), point OperatorHub at the mirrored catalog. Apply after cluster is up:

```
apiVersion: operators.coreos.com/v1alpha1
kind: CatalogSource
metadata:
  name: redhat-operator-index-mirror
  namespace: openshift-marketplace
spec:
  sourceType: grpc
  image: mirror.example.local:8443/ocp4/openshift4/redhat/redhat-operator-index:v4.20   #
  displayName: Red Hat Operators (Mirrored)
```

```
  publisher: Mirror-Local
  updateStrategy:
    registryPoll:
      interval: 30m
```

```
oc apply -f catalogsource-mirror.yaml

# Optional: disable the default (internet-facing) sources so nothing tries to reach the i
oc patch operatorhub.config.openshift.io/cluster --type merge \
  -p '{"spec":{"disableAllDefaultSources":true}}'
```

If your `oc-mirror` run produced an `ImageDigestMirrorSet` (`imageContentSourcePolicy.yaml` in older tooling) for **operator** images separate from what's already in `install-config.yaml` , apply that too:

```
oc apply -f oc-mirror-workspace/results-*/imageDigestMirrorSet.yaml
```

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10. Post-install verification checklist

```
export KUBECONFIG=~/ocp-install-run/auth/kubeconfig

oc get nodes                          # all 5 nodes Ready
oc get co                             # all ClusterOperators Available=True, Degraded=False
oc get pods -n openshift-vsphere-infra -o wide   # keepalived static pods, confirm 1 on A
oc get csr | grep Pending             # approve any pending node CSRs
oc get catalogsource -n openshift-marketplace
oc get clusterversion                 # confirms disconnected upgrade channel behaves (won
```

Sanity checks specific to "no external LB":

```
# From the bastion / any client with the DNS records
curl -sk https://api.ocp1.example.local:6443/version         # should return API version
curl -sk https://console-openshift-console.apps.ocp1.example.local -o /dev/null -w '%{htt
```

If either fails but nodes are Ready, re-check §2 (port group) and NSX rule 16 (VRRP protocol 112) first — that's the failure mode unique to the no-external-LB design.

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11. Quick troubleshooting map

Symptom	Likely cause
Bootstrap never completes, nodes can't	Mirror registry unreachable or CA not trusted → recheck

pull images	NSX rule 3 + `additionalTrustBundle`
`api.<cluster>` doesn't resolve/connect but nodes are Ready	DNS record missing/wrong, or VIP not up → check §1 and §2
VIP assigned but flaps between nodes constantly	NSX blocking VRRP (protocol 112) between control-plane nodes
`oc get pods -n openshift-vsphere-infra` shows keepalived CrashLoopBackOff	Port group security settings (§2) not applied
etcd health/leader election flapping	NTP not configured/reachable — apply §8
Operators fail to install from OperatorHub	CatalogSource image tag doesn't match what was actually mirrored — verify with `oc-mirror` results manifest

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Replace every placeholder (`<...>` , example IPs/hostnames) with your actual environment values before applying. Keep `install-config.yaml` , the mirror CA, and the pull secret out of version control or store them in a secrets manager.